From Molecular Redox Motifs to Polymer Grafts: Attachment Thermodynamics and Neighboring-Site Effects

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Internal working draft. Pending calculations and validation are identified explicitly.

Abstract

A favorable molecular redox potential does not establish that a motif can be attached to a polymer or retain useful behavior among neighboring grafts. We develop an exact-identity framework that connects molecular redox calculations to defined attachment reactions and finite polymer occupancy cycles. Its central comparison uses balanced free-energy differences to distinguish the effect of a shared base, the change from a capped site to an explicit polymer fragment, and the response to a neighboring graft. A retrospective benchmark of 39 potentials in acetonitrile gives a mean absolute error of 0.247 V, setting the accuracy context for molecular screening. In an exact PMDI example, first-site grafting to a five-repeat model is 5.3–6.1 kJ mol^{-1} less favorable than grafting to the local capped model; common precursor and base terms cancel from this difference. A matched adjacent-site cycle gives a nonadditivity of -1.86 kJ mol^{-1} for sampled minima with separated cores. Phthalimide controls show that an intervening ungrafted repeat can still permit close-core local minima, while scaffold mass changes the interpretation of nominal molecular capacity. These results establish how reaction definition, polymer context and occupancy enter candidate assessment without treating them as interchangeable descriptors. Absolute grafting signs remain conditional on ion speciation, and broader conformer convergence, electronic coupling and material accessibility remain necessary before synthesis selection.

Keywords: redox-active polymers; molecular screening; grafting thermodynamics; molecular identity; conformer ensembles; acetonitrile; density functional theory; chemical provenance; multiobjective selection.

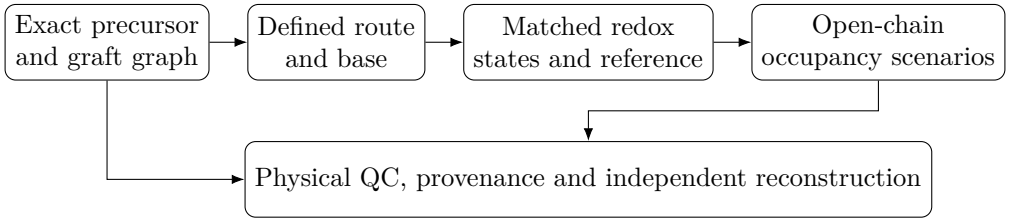


Figure 1: Workflow overview. Molecular redox properties, route-dependent grafting thermodynamics and neighboring-site effects are separate outputs linked by exact chemical identity. Execution correctness is evaluated independently of experimental accuracy. Every selection criterion is assessed separately for the same exact graft identity and declared conditions.

1 Introduction

Redox-active polymers and colloids provide charge-storage architectures that can reduce transport of active groups through separators. Aqueous polymer flow batteries and nonaqueous redox-active colloids establish distinct realizations of that approach.[1, 2] Kim and co-workers further demonstrated redox-matched insoluble polymer beads coupled to soluble mediators.[3] For an already formed support, selecting a redox group requires both a useful electrochemical response and a feasible reaction at an accessible polymer site.

Post-functionalization of chloromethylated polystyrene is also an established route to oxidation-active materials. Oh Biying and co-workers attached phenothiazine and TEMPO groups to partially chloromethylated commodity polystyrene and evaluated the resulting polymers as battery electrodes.[4] This precedent motivates evaluating proposed grafts across attachment chemistry and polymer context, with reactive-handle density distinguished from conversion and spatial contact.

High-throughput studies have explored quinone redox couples, while D3TaLES and RedDB organize redox-active chemical space for nonaqueous and aqueous electrolyte studies, respectively.[5–7] Such efforts motivate systematic design, while computational electrochemistry identifies reference, solvation and charge-state errors that complicate transferring potentials between chemical environments.[8–10] A potential of a free parent molecule cannot alone establish the behavior of an exact graft. Attachment introduces precursor access, selectivity and reaction conditions; neighboring grafts introduce additional conformational and electrostatic response. Reactive-handle density, conversion, sequence spacing and spatial contact must therefore remain distinct.

Self-association is already a demonstrated design concern for diimide electrolytes. Wiberg and co-workers combined aqueous NDI flow-battery experiments with charge-state-dependent dimer calculations, identifying electrolyte composition and steric substitution as relevant design variables.[11] These findings motivate evaluating neighboring-site effects within a declared molecular and electrolyte context.

Multistep redox screening has already been developed for organic radical polymer anodes and flow-battery anolytes. Achazi and co-workers combine conformer sampling with successive electronic-structure filters and benchmarked redox calculations.[12] General workflow systems likewise provide provenance and robust execution for large quantum datasets.[13–15] These advances establish the computational foundation for molecular selection. For post-functionalization of a preformed support, an additional question is how a selected molecular structure connects to an accessible precursor, a defined attachment reaction and a local polymer environment.

Here we address that connection through balanced thermodynamic comparisons on exact graft identities (Figure 1). We distinguish three effects: a base-dependent contribution shared by reactions with the same stoichiometry, the change in first-site attachment between a capped reference and an explicit open oligomer, and the change in attachment caused by a neighboring graft. The latter two comparisons cancel common reagent terms and test polymer context directly within a specified set of conformers. Applying them to an enumerated PMDI design links molecular screening to an explicit polymer feasibility question, while phthalimide controls test whether sequence spacing prevents spatial contact. The contribution is this chemically matched comparison across molecular, reaction and occupancy models; the present evidence does not establish a globally optimized polymer or a complete material shortlist.

2 Methods

2.1 Molecular identity, candidate generation and attachment routes

We model post-functionalization of a preformed polymer bearing reactive benzyl chloride sites. A 4-methylbenzyl chloride fragment provides a local capped-site reference. Its grafted products represent a polymer attachment site; the full support, crosslinks and graft distribution require separate polymer models.

Current molecular graph processing uses RDKit 2026.3.5.[16] Systematic designs vary declared viologen, pyromellitic-diimide and naphthalene-diimide cores, attachment linkers, terminal substituents and individual ring substitutions. Exact stereochemical graphs and precursor/product pairs are deduplicated, while every generating recipe is retained. Charge states and multiplicity hints are checked for electron-count consistency. A declared redox ladder remains a computational hypothesis until its states pass the appropriate property-specific checks. Molecular weight and nominal gravimetric capacity, $Q = nF/(3.6M)$, are bookkeeping descriptors; they do not establish accessible electron count, polymer capacity or synthetic feasibility.

The attachment model specifies a charge- and atom-balanced precursor/product pair and a uniquely mapped new heavy-atom bond from each verified upstream design. Evidence is bound to that exact identity, so a precedent for a different product cannot authorize the design. A prospective route gate requires documented precursor access, attachment selectivity, defined reaction conditions, a balanced route, redox-core retention and source evidence. It distinguishes proceed, hold and exclusion decisions. Family-level precedents are retained with their scope and are not automatically promoted to exact-candidate synthesis evidence. Historical calculated properties and explicitly scoped method controls remain available even when a prospective synthesis decision is unresolved.

2.2 Admission across all required dimensions

A candidate is eligible for ranking only after every required criterion passes in its declared context. The criteria cover operating redox window, material capacity, state stability, intended charge transfer, practical attachment, defined-route thermodynamics, unwanted motif association, unwanted intersite electronic coupling, occupancy sensitivity and polymer accessibility. A failed criterion excludes the design within that route or operating context. Missing evidence or an undeclared acceptance threshold leaves it pending. Route calculations use a reaction context; operating properties share a separately declared material and electrolyte context. Exact product identity is required across the joined records.

For a numerical criterion, the entire qualified uncertainty or scenario interval must lie within the accepted bounds. An interval crossing a boundary remains unresolved. Parent-benchmark residuals are not assigned as candidate-specific prediction intervals. The current material-capacity, charge-transfer, motif-association and intersite-coupling cutoffs remain undeclared; the model operating window and nonpositive defined-route free-energy screen are explicit conventions rather than experimentally validated universal thresholds. Pareto comparison is restricted to candidates that pass every criterion, with the two electrode pools treated separately. No weighted score supplies a missing pass.

2.3 Electronic structure and solution free energies

Geometry-level and native-Hessian calculations use r^{2}SCAN with D4 dispersion and SMD acetonitrile [17–19]. The geometry-level basis is def2-SVP for neutral and cationic states and def2-SVPD

for anions. Active electronic energies use ωB97M-V/def2-TZVPD on the selected geometry, with the same solvent model.[20–22] Density fitting uses the implementation-default auxiliary bases with `only_dfj=False`; the hybrid calculations therefore fit both Coulomb and exact-exchange contributions. The stored calculations use PySCF 2.14.0 and GPU4PySCF 1.8.1.[23, 24] The SI describes implementation-level numerical controls and the remaining release qualification. Supporting electrolyte ions are not implicitly introduced into these bare-solute calculations. Explicit ion-pair models are identified separately.

The reported molecular free-energy model is

$$G = E_{\mathrm{SMD}}^{\omega\mathrm{B97M-V}} + G_{\mathrm{nuclear}}^{\mathrm{qRRHO}} - RT\ln(2S+1). \tag{1}$$

The nuclear partition uses the solution geometry-level DFT Hessian. Rigid translations and rotations are excluded by a mass-weighted geometric projection, rather than by deleting all frequencies below a numerical threshold. The xTB 6.7.1 thermodynamic kernel is used only as a qRRHO integrator for the explicit DFT internal-mode list.[25, 26] Frequencies are unscaled. Modes between -5 and 0 cm^{-1} are accepted only within the stated minimum tolerance and enter the partition as $|\nu|$; more negative modes fail. The rotational symmetry number is determined by xTB from the exact geometry. The primary rotor cutoff is 50 cm^{-1}. Cutoffs of 25 and 100 cm^{-1} provide sensitivity estimates. This use of the thermodynamic kernel does not substitute a semiempirical Hessian for a DFT Hessian. The spin factor assumes an unsplit ground multiplet; unresolved state ordering remains a separate admission issue.

Thermodynamic acceptance requires the largest absolute Cartesian gradient component no larger than 4.5×10^{-4} hartree bohr^{-1}, Hessian asymmetry no larger than 10^{-6} in the stored atomic-unit matrix, and a lowest internal frequency no smaller than -5 cm^{-1}. The complete Hessian columns for atom 0/y and the first oxygen/z coordinate are tested at 0.0005 and 0.001 bohr; iron or the final atom is used when oxygen is absent. Each sampled column must meet a 1% relative-norm error threshold. This tests two columns, not the full Hessian. The absolute spin-contamination threshold is $|\langle S^2\rangle - S(S+1)| \leq 0.1$. Full integration-grid response, finite numerical values and spin diagnostics are checked separately. SCF convergence alone is insufficient for thermodynamic admission. Missing thermal or electronic quantities remain missing. Geometry optimization uses geomeTRIC.[27] The charge-dependent geometry/Hessian basis is a protocol imbalance. Five benchmark neutral partners were separately reoptimized with def2-SVPD and evaluated with a derivative-checked Hessian at that level. Their actual alternative protocol identities remain separate from the primary certificate. Existing anion and Fc records are held fixed to separate nuclear and geometry-induced active-energy effects; this is a sensitivity test rather than complete basis convergence.

2.4 Directional solution relaxation

For the route-screened TEMPO comparison, we evaluate state-specific frozen-slow continuum response in MeCN using a Pekar partition of SMD electrostatics.[28] All six equilibrium and vertical calculations share the active electronic level and basis; fast polarization relaxes while the initial slow polarization remains fixed. The resulting directional electronic relaxation penalties use the two checked geometry-level minima. They contain no nuclear thermal correction and are separate from gas inner-sphere or Born outer-sphere estimates. The SI gives the energy cycles, solvent constants and model limitations.

2.5 Redox reference and retrospective accuracy assessment

For a one-electron oxidized/reduced pair, potentials are calculated from $G_{\mathrm{ox}} - G_{\mathrm{red}}$ and referenced to a matched ferrocene/ferrocenium calculation. The experimental comparison uses exact free-parent identities and documented acetonitrile measurements with direct or source-defined reference conversion. Lower-tier measurements and excluded events are retained in the source catalog.

The frozen retrospective panel contains 39 potential endpoints from 30 molecules. Primary study locations and reference handling for all seven studies are tabulated in the Supporting Information. It includes all currently sourced strict endpoints and is not a held-out test set. Independent reviewers reconstruct state free energies, the reference and potential differences from backing records. Statistics use unrounded calculated potentials, retaining family, redox-step and original-study labels. Molecule- and study-cluster bootstrap summaries use 2,000 replicates and seed 20261021; intervals require at least five clusters. These summaries describe this panel and are not calibrated prediction intervals for new structures. The primary statistics use no fitted offset or outlier removal. Post hoc endpoint-, molecule- and study-held-out offsets are reported separately, using either a single training-set mean residual or separate means for the two redox steps. All endpoints of a held-out molecule or study are excluded together. Because the residual structure motivated these analyses, they are internal diagnostics rather than prospective validation.

2.6 Defined grafting reactions and standard states

For an N–H or O–H precursor, the local reference reaction is

$$\mathrm{RCl} + \mathrm{HNu} \rightarrow \mathrm{RNu} + \mathrm{HCl}. \tag{2}$$

Pyridyl quaternization is treated as a distinct reaction class with the appropriate precursor and chloride charge. The neutral-substitution reference is not interpreted as the actual free energy of a synthesis that includes a base. For a defined base B, we additionally evaluate

$$\mathrm{RCl} + \mathrm{HNu} + \mathrm{B} \rightarrow \mathrm{RNu} + \mathrm{BH^+} + \mathrm{Cl^-}, \tag{3}$$

$$\mathrm{RCl} + \mathrm{HNu} + \mathrm{B} \rightarrow \mathrm{RNu} + [\mathrm{BH^+} \cdots \mathrm{Cl^-}]. \tag{4}$$

The separated-ion and contact-pair representations are alternative thermodynamic models, rather than an assumed equilibrium population or a precipitation model. Each reaction is checked for atom and charge balance.

The native thermal translation convention is ideal-gas 1 atm. Reaction sums are converted to a 1 M solute standard state by adding $\Delta n RT \ln(RTc^\circ/p^\circ)$. The correction cancels for the neutral reference and separated-ion routes and is approximately $-7.93\ \mathrm{kJ\,mol^{-1}}$ for the contact-pair route at 298.15 K. The DBU/DIPEA pilot computes each neutral base, protonated base and protonated-base/chloride contact pair. Three geometry seeds are evaluated per new species, and thermochemistry is applied to the selected electronic-energy minimum. Reused component values retain their original source hashes. The difference between paired and separated representations is a model sensitivity, not an uncertainty interval.

A separate external term check compares $\mathrm{B} + \mathrm{HCl} \rightarrow \mathrm{BH^+} + \mathrm{Cl^-}$ with the author-maintained MeCN basicity collection and recommended acid table.[29–31] The conversion is $\Delta G^\circ = -RT \ln(10)[\mathrm{p}K_a(\mathrm{BH^+}) - \mathrm{p}K_a(\mathrm{HCl})]$. Base identity is matched by the exact neutral molecular graph; analogues are not substituted. This comparison uses 298.15 K, while exact source temperature and activity conventions remain under review. HCl cancels from the full all-DFT grafting cycle. Replacing only its capture term therefore tests sensitivity to a term and shared-error cancellation; it does not calibrate the complete reaction.

2.7 Finite-polymer construction and conditional occupancy cycles

Oligomer construction preserves an open carbon backbone, the intended graft bond and declared stereochemistry. Monomers, dimers and trimers are included in the original interaction panel; no tetramers are included. GFN2-xTB/ALPB(acetonitrile) warm starts provide conformational seeds for subsequent DFT evaluation.[32, 33] Starting pose labels do not determine the final contact class. XYZ audits independently inspect composition, bonding, stereochemistry and severe clashes.

The matched spacing control uses a five-repeat backbone with reactive sites 2, 3 and 4 and plain styrene end-padding units at positions 1 and 5. The six occupancy graphs are empty, singly grafted at sites 2, 3 or 4, and doubly grafted at 2/3 or 2/4. Unreacted benzyl chloride and unfunctionalized styrene units remain chemically distinct. The graft is a neutral phthalimide method control. The initial search uses seed 20261020, four embedded structures and two xTB seeds per starting mode, with two selected DFT poses per graph giving twelve electronic calculations. Unbiased starts are supplemented by open and slipped starts for double grafts. One declared backbone configuration is used; it is not a model of an atactic sequence ensemble. Fully occupied short oligomers represent dense local scenarios; these calculations assign no experimental loading distribution or occupancy probability.

The neighboring-site free-energy nonadditivity is

$$\Delta\Delta G_{ij} = G_{ij} + G_{00} - G_i - G_j. \tag{5}$$

This quantity is also the difference between grafting the second site in the presence and absence of the first graft when the same reagent and byproduct terms are used. It includes changes in the backbone, conformation and solvent response and is not a pure pendant-core pair interaction. Electronic $\Delta\Delta E$ and thermodynamic $\Delta\Delta G$ are kept separate. An absolute conditional grafting free energy additionally includes the defined base and byproduct terms.

For the exact PMDI example, first attachment to the empty chain P_0 is also compared with attachment to the local benzyl-chloride model RCl. Writing the corresponding products as P_i and RM, the local-to-polymer difference is

$$\delta G_{i,\mathrm{poly-local}} = G(P_i) + G(RCl) - G(P_0) - G(RM). \tag{6}$$

The same precursor, base and byproduct terms cancel from this balanced difference. It measures the change between the two specified molecular contexts, including their geometries and solvent response. It is distinct from the neighboring-graft cycle in Eq. 5.

For separate, intact molecular segments, frozen-fragment energies distinguish interaction from deformation:

$$E_{\mathrm{int}} = E_{AB} - E_A^{AB} - E_B^{AB}, \tag{7}$$

$$E_{\mathrm{def}} = (E_A^{AB} - E_A^0) + (E_B^{AB} - E_B^0). \tag{8}$$

Their sum equals the segment association energy referenced to relaxed isolated segments. This partition avoids cutting covalent bonds or adding new caps. It nevertheless contains the full segments, is not counterpoise corrected, and does not establish an association free energy or electronic coupling.

2.8 Evidence and execution controls

Every reported field points to exact inputs, numerical settings and immutable backing records. Physical checks govern acceptance separately from scheduling success. Durable geometry checkpoints remain unqualified until normal convergence, topology, spin and thermodynamic checks pass.

Independent GPU reservations and bounded CPU thread pools govern parallel calculation. Numerical replay, synthetic throughput, CPU concurrency and actual quantum restart controls are reported in the Supporting Information; these test execution or numerical equivalence rather than experimental accuracy.

3 Results

3.1 Systematic designs and benchmark coverage

The declared enumeration generates 1,593 recipes and 837 distinct designs: 459 viologens, 243 NDI derivatives and 135 PMDI derivatives. This pool concentrates on reduction-active motifs and includes two related diimide cores. BQ/AQ and oxidation-active chemistry have historical examples but lack equivalent systematic coverage; shielding and linker rigidity also need deliberate matched comparisons. These boundaries define a focused study rather than a comprehensive chemical-space search.

The historical evaluation comprises 75 candidates and the same 75 reaction identities, with 72 unique precursors. Exact graph matches can reuse calculations with matching state and protocol provenance. The prespecified all-axis assessment combines this cohort with the systematic enumeration and one method control, giving 879 distinct graphs and 878 candidate designs. All 878 remain pending under that assessment profile; incomplete criteria are distinguished from demonstrated chemical failure. A broader maintained identity registry includes later proposals and registered local models, giving 904 distinct graft-site graphs. The exact PMDI-ethyl and baseline viologen examples below link these identities to route and calculation evidence without inferring a complete material shortlist from molecular descriptors.

The retrospective free-parent benchmark includes 39/39 sourced potentials from 30 molecules and seven studies, with MAE 0.247 V, RMSE 0.279 V and bias +0.190 V (Figure 2). Molecule-cluster 95% bootstrap intervals are [0.213,0.286], [0.245,0.313] and [0.128,0.257] V, respectively. Seven unequal study clusters provide weakly calibrated intervals, reported separately in the SI. An independent reconstruction contains 111 checks with no discrepancies. The final PMDI second-reduction endpoint is included; its alternative triplet lies 0.707 eV above the selected singlet electronically, while the alternative thermal spin gap remains unresolved.

Coverage differs materially from the design space. Twenty-two endpoints are quinones, which are absent from the systematic enumeration. The imide subset contains nine endpoints from six molecules in one study, including three PMDI derivatives and three phthalimides; viologen contributes one molecule, and there is no NDI endpoint. These results define accuracy for the measured free parents and do not validate 837 grafted designs or polymer environments.

3.2 Reference bias and charge-state errors are distinguishable

All 30 first-step errors are positive, with mean +0.266 V. The nine second-step errors have mean −0.064 V and MAE 0.185 V. Their lower MAE can reflect cancellation between systematic contributions and should not be read as superior second-reduction accuracy. Squared bias accounts for approximately 46% of the overall mean-squared error. A molecule-held-out constant offset reduces MAE/RMSE to 0.158/0.211 V; separate first- and second-step offsets give 0.106/0.166 V. Study-held-out values are 0.182/0.232 and 0.124/0.177 V, respectively (SI). These internal analyses diagnose structure in the residuals while leaving the primary predictions unchanged.

The computed Fc absolute potential is 4.379 V. Stored comparison values, 4.618–4.662 V, use other functionals and cannot be adopted as level-matched replacements. Their difference from the

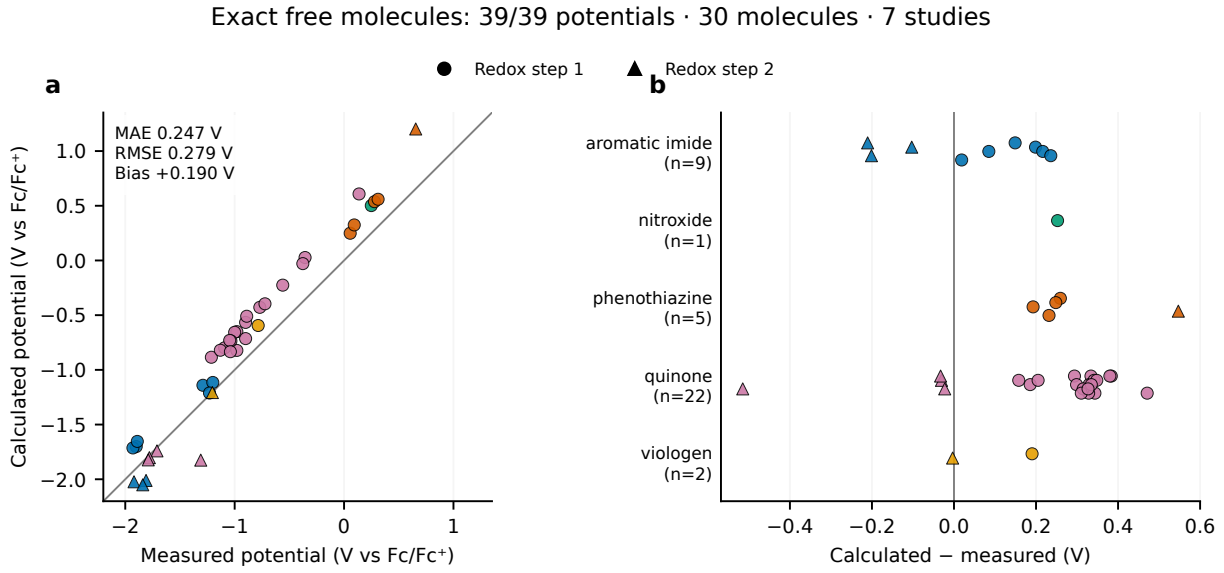


Figure 2: Unfitted retrospective benchmark for 39 exact free-parent potentials in MeCN. Family and redox-step labels expose unequal coverage. All sourced endpoints and outliers are retained; no fitted offset is applied.

present reference is similar to the positive first-step bias, making a reference contribution plausible. Multiple organic-reference differences that cancel Fc are examined in the SI. Fixed-geometry B3LYP and PBE0 reference controls shift the electronic gap downward and worsen the reference-only counterfactual bias (SI); they do not establish alternative absolute potentials. Nevertheless, all nine paired step spacings are too large in magnitude: a common reference shift cancels from a spacing and cannot explain this result. Reference, charge-state solvation and ion-pairing errors can coexist. A large quinone study found distinct first/second-reduction intercepts under other methods in DMF versus SCE; it provides relevant precedent, not a numerical calibration for MeCN/Fc.[10] In five neutral-SVPD controls, changing both geometry and Hessian basis lowers the first potential by 11.7–35.1 mV when the original anion and Fc are held fixed. Nuclear contributions are 7.8–21.3 meV; geometry-induced active-energy contributions are 0.4–13.8 meV (SI). The imbalance is measurable but does not explain the full observed bias. A spin-only partition also omits ferrocenium orbital/vibronic contributions; their sign and limiting size are discussed in the SI.[34] Changing the nuclear rotor cutoff over 25, 50 and 100 cm^{-1} gives a maximum benchmark endpoint span of 25.4 mV and aggregate MAE values of 0.245–0.247 V (SI). The primary 50 cm^{-1} convention is retained.

3.3 Route-admitted designs connect enumeration to molecular calculations

The baseline N-methyl viologen in the enumerated pool maps exactly to the N-(4-methylbenzyl) local graft. Its remaining pyridyl nitrogen and mono-methyl bipyridinium precursor match a reported attachment to preformed Merrifield beads in refluxing MeCN.[3] This quaternization passes the prospective route screen and releases chloride without a proton-capture term. Native calculations for the declared dication singlet, radical-cation doublet and neutral singlet give unfitted first and second reduction potentials of −0.608 and −1.208 V versus the matched Fc reference. The 0.600

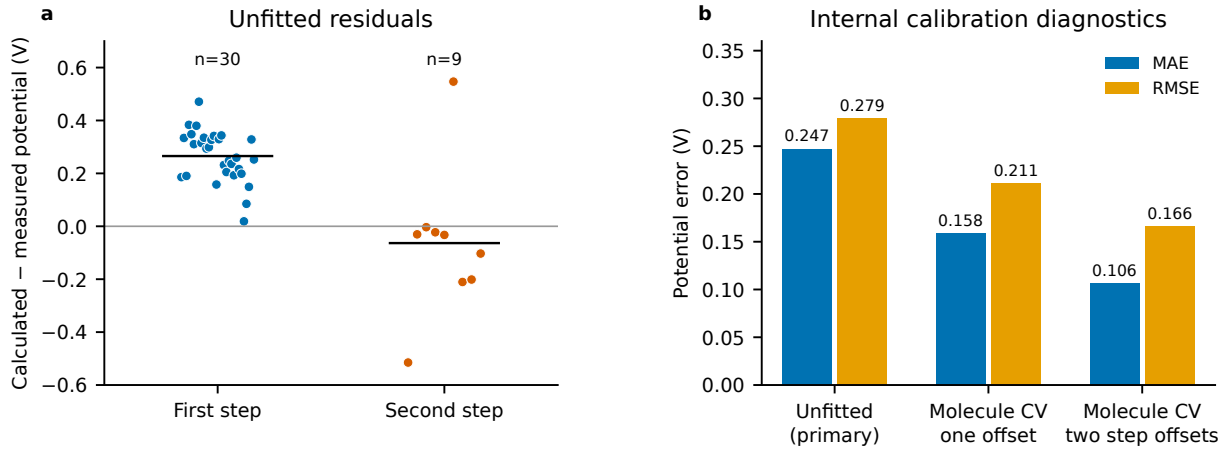


Figure 3: Unfitted step-dependent residuals and internal molecule-held-out offset diagnostics. Training offsets exclude the entire target molecule. Diagnostic calibration is shown separately from the primary benchmark and is not an external test on new compounds.

V spacing corresponds to +57.91 kJ mol^{-1} for disproportionation of two isolated radical cations. Source geometries that failed the native minimum criterion were repaired before thermodynamic acceptance (SI). This monomer comparison does not test radical dimerization, polymer-mediated interactions or degradation.

For this same exact graft, the separated-ion quaternization model gives $\Delta G = -11.49$ kJ mol^{-1} in MeCN at 298.15 K. The mono-methyl bipyridinium precursor was computed independently from three fixed starting conformers; the lowest precursor G at the primary rotor setting defines the reported cycle. The 25–100 cm^{-1} settings give -13.30 to -11.49 kJ mol^{-1}. This cycle links the enumerated identity, reviewed attachment route, redox ladder and grafting thermodynamics without a base-capture offset. It represents a capped site with separated chloride: ion association, counterion exchange, the experimental reflux temperature and polymer constraints are not included. The small-anion solvation uncertainty therefore remains relevant to its sign and cannot be calibrated by transferring the acid–base correction to this different reaction.

As an oxidation-active comparison outside the enumerated pool, the existing 4-oxy-TEMPO local graft has a direct attachment precedent on preformed Merrifield resin. A patent reports reaction of 4-hydroxy-TEMPO with the resin using NaH in DMF or THF and reports N-oxyl content in the washed product.[35] The mapped local graft passes the route screen, and its independently reconstructed MeCN oxidation potential is +0.579 V versus Fc/Fc$^+$. A separately checked frozen-slow SMD calculation gives electronic solution-relaxation penalties of 1.477 eV for reduction and 1.639 eV for oxidation. Their 10.4% directional asymmetry exceeds the predeclared 5% diagnostic for an equal-curvature approximation, so no single symmetric parameter is qualified. These composite local-model descriptors do not establish measured rates.

The viologen cycle and TEMPO solution response establish molecular quantities. TEMPO grafting, both polymer assessments and viologen attachment under actual polymer conditions remain incomplete. Material decisions remain pending. The PMDI case below adds finite grafting and occupancy calculations while retaining its unresolved practical-route requirements.

3.4 A fixed base shifts grafting thermodynamics without changing ranking

Of 75 local reference reactions, 67 have complete free energies and eight remain held. The base pilot combines five complete candidate reactions with six checked DBU/DIPEA components to give twenty base-assisted values. For a fixed base and ion representation,

$$\Delta G_{\mathrm{graft,B}} = \Delta G_{\mathrm{graft,HCl}} + \Delta G(\mathrm{B} + \mathrm{HCl} \rightarrow \mathrm{BH^+} + \mathrm{Cl^-}), \tag{9}$$

with the paired analogue and its standard-state correction defined separately. The separated-ion constants are $-102.78\ \mathrm{kJ\,mol^{-1}}$ for DBU and $-70.98\ \mathrm{kJ\,mol^{-1}}$ for DIPEA. Thus the base choice can change signs but cannot reorder candidates within a given representation.

The two benzoquinone HCl-reference values, $+57.63$ and $+56.03\ \mathrm{kJ\,mol^{-1}}$, become $-45.14/-46.75\ \mathrm{kJ\,mol^{-1}}$ with DBU and $-13.34/-14.94\ \mathrm{kJ\,mol^{-1}}$ with DIPEA. AQ-1-oxy and NDI also change sign in this panel (Figure 4); this is not a benzoquinone-specific rescue. These values motivate testing a defined base-assisted route, while selectivity, precursor access, salt speciation and support accessibility remain separate route gates.

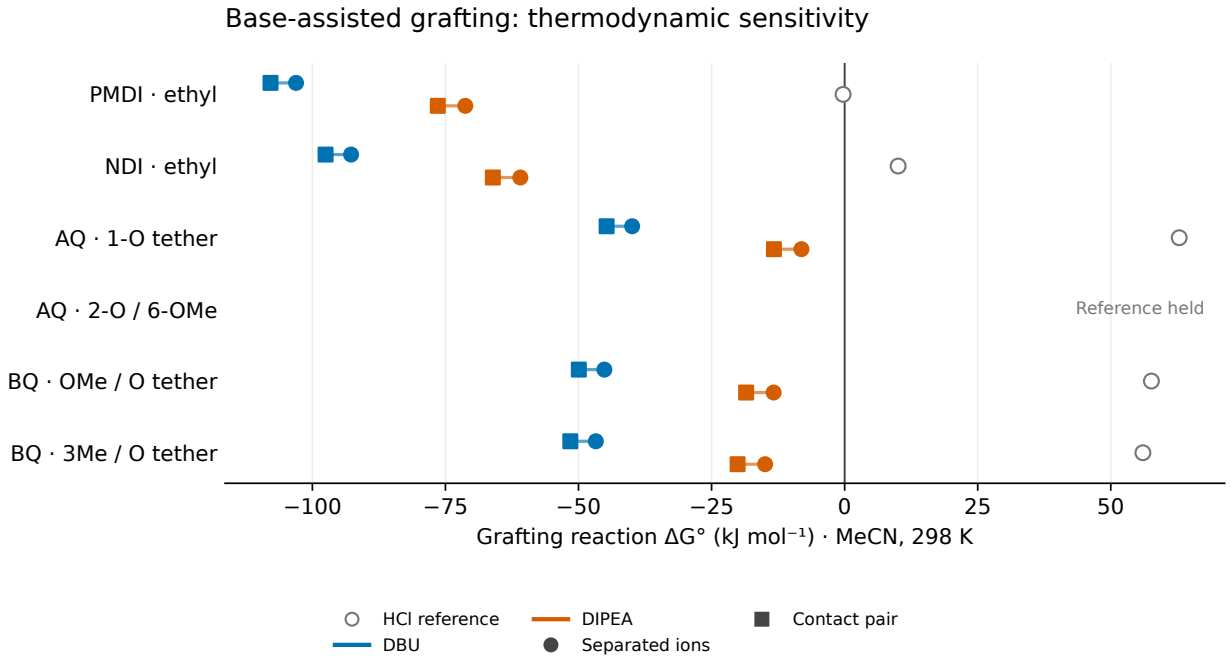


Figure 4: Candidate HCl-reference reactions plus a constant per-base proton-capture term. Separated ions and contact pairs are alternative thermodynamic representations; their differences do not establish measured ion-pair populations.

The revised MeCN collection gives pKaH(DBU)=24.31 and the recommended acid table gives pKa(HCl)=10.30.[29–31] At 298.15 K these imply $-79.97\ \mathrm{kJ\,mol^{-1}}$ for capture, versus $-102.78\ \mathrm{kJ\,mol^{-1}}$ computed, a discrepancy of $-22.81\ \mathrm{kJ\,mol^{-1}}$. Replacing only this term leaves the two DBU/BQ sums negative ($-22.33/-23.94\ \mathrm{kJ\,mol^{-1}}$), but changes shared-error cancellation because HCl cancels from the full DFT cycle. These hybrid sums are sensitivities, not calibrated grafting predictions. Two additional HCl/Cl-free relative-basicity reactions, for pyridine and triethylamine against DBU, differ from the tabulated scale by $+4.58$ and $+1.40\ \mathrm{kJ\,mol^{-1}}$ (SI). For DIPEA, spectrophotometric titration reports $\log K^{\mathrm{H}} = 18.64(1)$ in MeCN at 25 °C, while a separate NMR

competition estimate for DIPEAH$^+$ hexafluorophosphate gives 18.7 in MeCN-d_3.[36, 37] Combining the former with the DBU scale implies +32.36 kJ mol^{-1} for the DIPEA/DBUH$^+$ proton-transfer reaction, compared with +31.80 kJ mol^{-1} computed. Combining it with the HCl table instead implies a capture term of −47.60 kJ mol^{-1}, versus −70.98 kJ mol^{-1} computed. Indicator calibration, ion association and activity conventions are not established to match across sources. These comparisons do not identify a unique erroneous species or validate the complete grafting reaction. The previously reported 18.6 value remains a separate PCM estimate.[38] DIPEA reaction signs remain conditional.

All 87 completed reference and base-assisted reaction values retain their nominal sign over those cutoff scenarios; the largest span is 4.46 kJ mol^{-1}(SI). This settings test leaves the independent route-validation requirements unchanged.

3.5 An exact PMDI design links molecular, grafting and occupancy assessments

The enumerated PMDI-ethyl design maps exactly to the mono-N-ethyl/NH precursor and its N-(4-methylbenzyl) graft-site model. Its checked molecular potentials are −1.076 and −1.977 V versus the computed Fc reference. The HCl-reference grafting value is −0.29 kJ mol^{-1}; including the defined DBU/separated-ion term gives −103.07 kJ mol^{-1}. These values provide molecular and reaction-model evidence, with the benchmark and proton-transfer limitations described above.

Example 19 of EP0010440B1 reports isolation of the exact mono-N-ethyl/NH PMDI precursor with a stated NMR assignment.[39] Molecular N-alkylation of PMDI in the presence of CsF has also been reported, with crystallographic characterization of an N,N′-diethyl derivative.[40] These reports support precursor access and the attachment reaction class, respectively. They do not establish DBU-assisted benzylation of the exact mono-N-ethyl precursor, selective attachment to a Merrifield-type bead or retention of redox performance after grafting. The exact route therefore remains pending for selectivity and core-retention evidence.

For the sampled singly grafted minima of the five-repeat model, first-site attachment gives ΔG = −97.7 and −97.0 kJ mol^{-1} at sites 2 and 3 under the defined DBU/separated-ion model. Relative to the capped reaction, the polymer-context penalties are +5.34 and +6.11 kJ mol^{-1}, respectively (Eq. 6). The common precursor and base terms cancel from these differences; the favorable absolute reaction signs retain the proton-transfer and speciation qualifications above. The penalties remain positive over the prescribed rotor-cutoff settings (Figure 5a). The small relative preference between the two sampled positions changes sign with that setting, so these single-conformer results do not establish a preferred graft site or a converged polymer equilibrium (SI).

The neutral capped molecule has a nominal two-electron capacity of 153.9 mAh g^{-1}. Retaining scaffold and residual chloride mass changes the composition-based ceiling: at three reactive handles per five repeats, one, two or three PMDI grafts give 61.2, 98.9 or 124.5 mAh g^{-1}, respectively (Supporting Information). These are conditional bounds for a dry neutral linear fragment with full utilization of an assumed two-electron budget. Crosslinker, electrolyte and charging counterion mass are excluded unless explicitly added. No graft-conversion distribution or accessible polymer capacity is inferred. An application target exceeding a scenario's ceiling would rule out that scenario; lying below the ceiling would still require independent evidence for utilization, stability and transport.

The completed paired-state calculation closes the four-state occupancy cycle for this exact design. At the primary rotor cutoff, $\Delta\Delta G$ = −1.86 kJ mol^{-1}: attachment at site 3 changes from −96.96 kJ mol^{-1} on the empty chain to −98.82 kJ mol^{-1} when site 2 is already grafted (Figure 6). Reversing the attachment order gives the same double-grafting free energy. The shared precursor,

base and byproduct terms cancel from the nonadditivity, which is identical for separated ions and the contact-pair representation. Values at 25, 50 and 100 cm^{-1} are −1.07, −1.86 and −1.82 kJ mol^{-1}, respectively.

The two PMDI cores are 16.30 Å apart in the paired minimum, despite occupying adjacent sequence sites (SI). This finite whole-chain response therefore does not measure a close-contact stacking interaction, isolated motif binding or electronic coupling. Each occupancy graph contributes one minimum. The example links enumeration, exact molecular response, route evidence and matched polymer thermodynamics while retaining a pending material decision: selectivity, broader conformer sampling and accessible utilization remain unresolved.

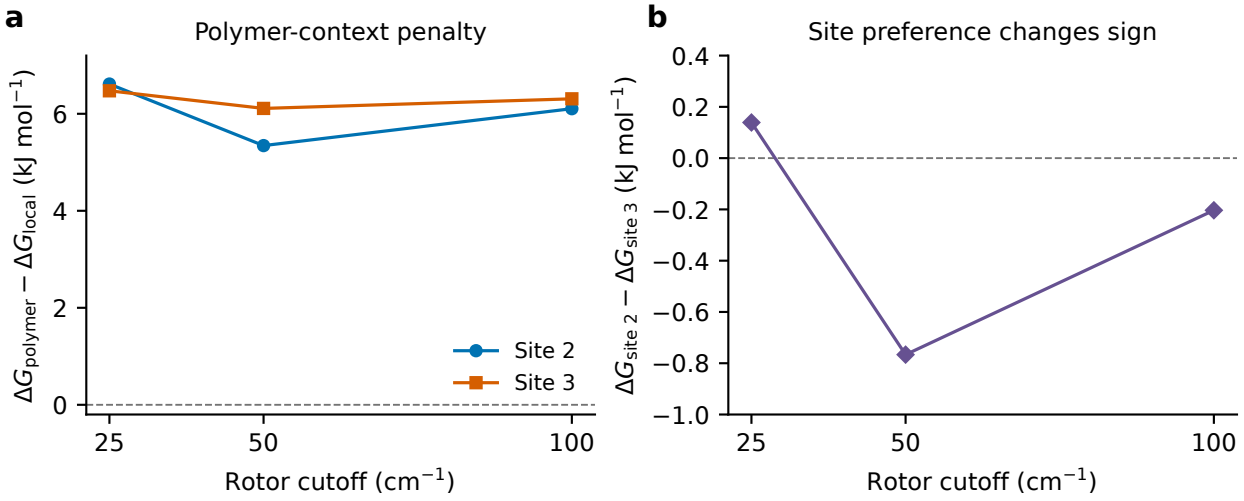


Figure 5: First-site PMDI grafting in sampled five-repeat polymer minima. (a) The polymer-minus-local free-energy penalty stays positive at each tested rotor cutoff; common precursor, base and byproduct terms cancel. (b) The small free-energy difference between grafts at sites 2 and 3 changes sign. Lines connect tested numerical settings and are not statistical uncertainty intervals. Each occupied graph contributes one minimum, so neither panel establishes a converged conformer ensemble or a preferred graft site.

3.6 Conditional grafting is demonstrated in a finite control

A five-repeat phthalimide model distinguishes reactive sites, plain-styrene end padding and occupied sites. All twelve electronic poses and their twelve native free energies pass their specified checks and independent reconstruction. A Boltzmann partition over the two known poses of each graph gives adjacent and one-spacer nonadditivities of −5.17 and −11.39 kJ mol^{-1}. These values include backbone, conformational and solvent response. They quantify conditional attachment within the declared set, rather than a pure motif-pair interaction.

The sequence interval does not fix spatial separation in these controls (Figure 7). Each occupied graph has both a close-core and a separated-core local minimum. Adjacent-site core centroids are 4.24 and 16.66 Å apart; the corresponding one-spacer distances are 4.77 and 20.20 Å. Within each sampled pair, the close-contact conformer has lower G by 11.9 and 7.1 kJ mol^{-1}, respectively. These same-graph differences include changes throughout the chain and solvent response, distinct from the four-state grafting nonadditivity. The one-spacer model therefore permits motif contact through backbone folding. The strength of this contact was assessed separately using frozen methyl-capped

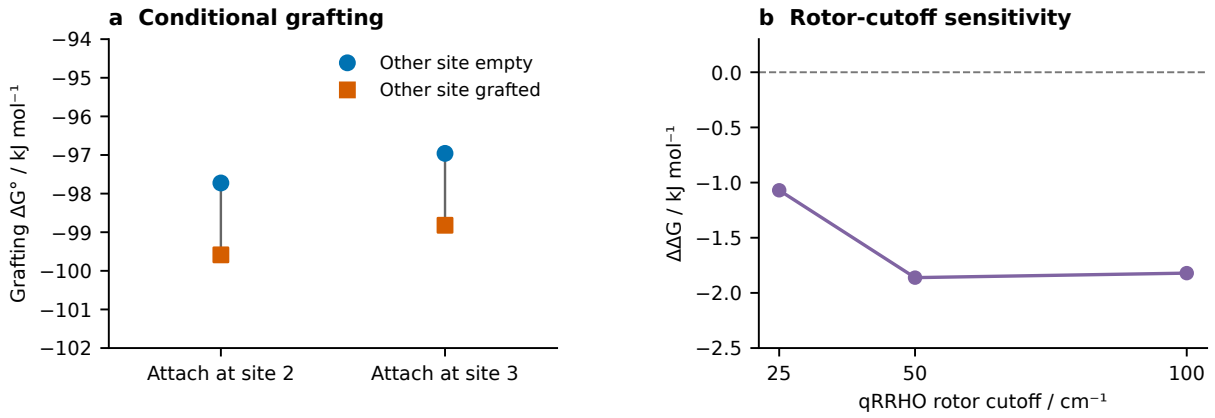


Figure 6: Finite PMDI neighboring-graft cycle in MeCN. (a) Defined DBU/separated-ion attachment free energies with the other site empty or grafted. Vertical segments show conditional differences for the same target site. (b) Nonadditivity at three rotor cutoffs; lines connect settings and are not uncertainty intervals. Common reaction terms cancel. All four graphs have one sampled minimum; the paired geometry has separated cores and does not establish stacking strength, electronic coupling or ensemble convergence.

phthalimide fragments. At the central cap length, the close adjacent and one-spacer poses give electronic interaction energies of -24.76 and -26.22 kJ mol^{-1} in the MeCN model, respectively; the separated poses give values near zero. These energies include cap and continuum response and lack basis-superposition correction. They are neither association free energies nor electronic couplings, and the four sampled poses do not establish contact populations or a spacing trend.

The maximum configurational lowering among these six two-pose sets is 0.14 kJ mol^{-1}. This small change describes only the two known conformers; it does not bound the error from undiscovered conformers. Changing the rotor cutoff gives cycle ranges of -5.17 to -4.64 and -12.34 to -11.39 kJ mol^{-1}. Broader sampling, multiple backbone configurations and DFT re-ranking remain necessary before interpreting a spacing design rule. Construction, conditional reactions and the finite-partition protocol are reported in the SI. The broader, separate smooth-surface MeCN control demonstrates why this assessment must be made for each cycle component. Both empty-backbone configurations meet the declared 10-to-20 input batch criterion: their finite-set ensemble free energies change by -1.03 and -1.23 kJ mol^{-1} at the primary partition settings (SI). The occupied sets remain incomplete, so these empty-component checks do not establish an occupied ensemble or a spacing trend. Alternate-model values do not enter the primary occupancy cycles.

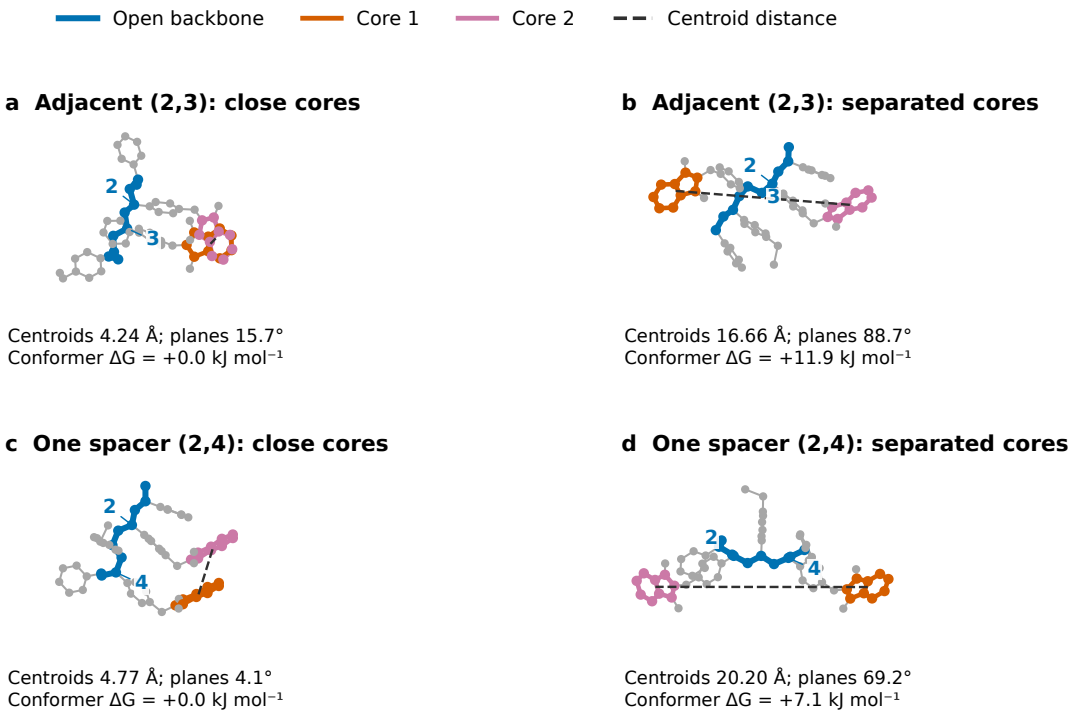


Figure 7: Accepted phthalimide control geometries at adjacent (a,b) and one-spacer (c,d) occupancy. Blue marks the open backbone; numbers identify occupied repeats. Orange and magenta mark conjugated core-ring atoms. Ungrafted site 3 in the one-spacer graph retains benzyl chloride; the end-padding repeats are plain styrene. Orthographic views use the saved XYZ coordinates, proper rotations and a common spatial scale; hydrogens are hidden and bond lines indicate checked connectivity. Dashed lines join core centroids, and angles refer to their fitted planes. Conformer ΔG is relative to the lower G in the same two-pose graph at 298.15 K and a 50 cm^{-1} rotor cutoff. These finite sets do not establish converged populations or isolated core-binding free energies.

4 Discussion

The balanced comparisons expose changes that a molecular-potential ranking cannot resolve. A fixed base contributes the same free-energy term to matched attachment reactions, whereas the first-site polymer penalty and neighboring-site nonadditivity arise from different molecular environments. Keeping the same exact graft identity across these comparisons makes their physical meaning explicit: the former measures replacement of the capped reference by an open polymer fragment, and the latter measures the response to adding a second graft. Neither quantity isolates pendant-core binding, because backbone deformation, conformation and solvent response remain included.

The PMDI example demonstrates this distinction. Favorable defined-route thermodynamics coexist with a measurable first-site context penalty and a small neighboring-site response in a conformation whose cores are separated. The phthalimide controls supply a complementary geometric observation: adding an ungrafted repeat does not preclude close-core local minima. Together, these findings motivate conditional occupancy calculations when experimental graft loading and sequence spacing are unknown. They do not assign a probability distribution to real grafting patterns. Converged ensembles and an intended charge-transport model are needed to turn these finite comparisons into a material design rule.

We prioritize weak unwanted association between neighboring grafts. Redox-polymer electrodes can also use electron hopping for charge propagation, with the mechanism depending on site concentration and chain mobility.[41] Fast bulk charge propagation has likewise been demonstrated in TEMPO polymer layers.[42] Experiments and molecular dynamics on soluble phthalimide polymers in DMF further show concentration-dependent transport and electrochemical kinetics, with similar behavior across the three backbones examined.[43] A material assessment therefore needs an intended charge-transport pathway and explicit competing processes. Neutral interaction energies and conditional grafting free energies do not determine transport or self-discharge rates.

The benchmark identifies a sizeable common bias and a separate step-dependent component. Reference cancellation, grouped cross-validation and paired spacings offer complementary diagnostics; none alone attributes error uniquely to Fc. Changing the fitted offset or charge-state protocol now would require a declared calibration set and a separate external test before claiming improved predictive accuracy.

The scope of the evidence determines how far these assessments transfer. The free-parent benchmark is concentrated in quinones and provides limited support for diimides and viologens. The basis controls quantify a specific protocol sensitivity rather than establish basis convergence. Base-assisted sums remain sensitive to continuum solvation and ion speciation, while the finite occupancy partition does not bound contributions from undiscovered conformers or an atactic polymer ensemble. For an exact candidate, these uncertainties accompany the chemical requirements for selective attachment, redox-core retention and accessible utilization. An all-criterion decision therefore requires candidate-specific evidence under a consistent set of operating and reaction conditions.

5 Conclusions

Selecting motifs for a preformed redox-active polymer requires a connection between the exact molecular graft, its attachment chemistry and its polymer environment. This work implements that connection through a benchmarked molecular calculation and balanced reaction and occupancy cycles. Common base terms cancel from within-base comparisons, while the exact PMDI case quantifies a first-site polymer penalty and a separate finite neighboring-graft response. The spacing

controls further show why a sequence spacer cannot be assumed to eliminate spatial contact. These comparisons provide a route-aware basis for evaluating proposed grafts and expose the evidence still required for a material decision. Their extension to synthesis priorities depends on matched solution speciation, broader conformer sampling, retained redox activity and accessible loading.

Supporting Information

Benchmark intervals and grouped offset diagnostics; relative-basicity controls; finite-polymer construction and conditional cycles; electronic fragment decomposition; numerical implementation, concurrency and quantum restart controls; evidence terminology.

Data and code availability

Versioned code, curated endpoint data and hash-verified evidence snapshots are available in the public project repository. A small ten-state CPU example includes the geometries, electronic energies, Hessians and thermochemical records needed to reconstruct the exact viologen reduction and local quaternization results with NumPy alone. Additional benchmark, grafting and finite-oligomer records are assembled in reviewer packages whose scope, locked environments and replay checks are described in the Supporting Information. Full raw-data archival deposition, its DOI and final deposition metadata remain outstanding.

Author contributions, funding and competing interests

These statements require confirmation by the authors and are intentionally not inferred in this working draft.

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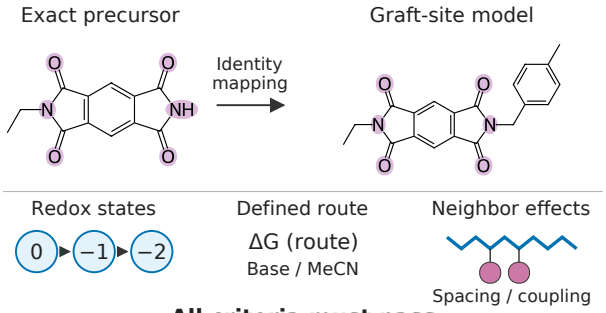
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